Note: I spent a good 90 minutes on this only for matlab to shit the bed and kill everything. I then redid everything in 20 minutes. If this prelab extra awful, then you know but I am done with this assignment. I will tank whatever grade this gets me.

SECTION 1

```
Fs = 8000;
T = 1/Fs;
```

SECTION 2

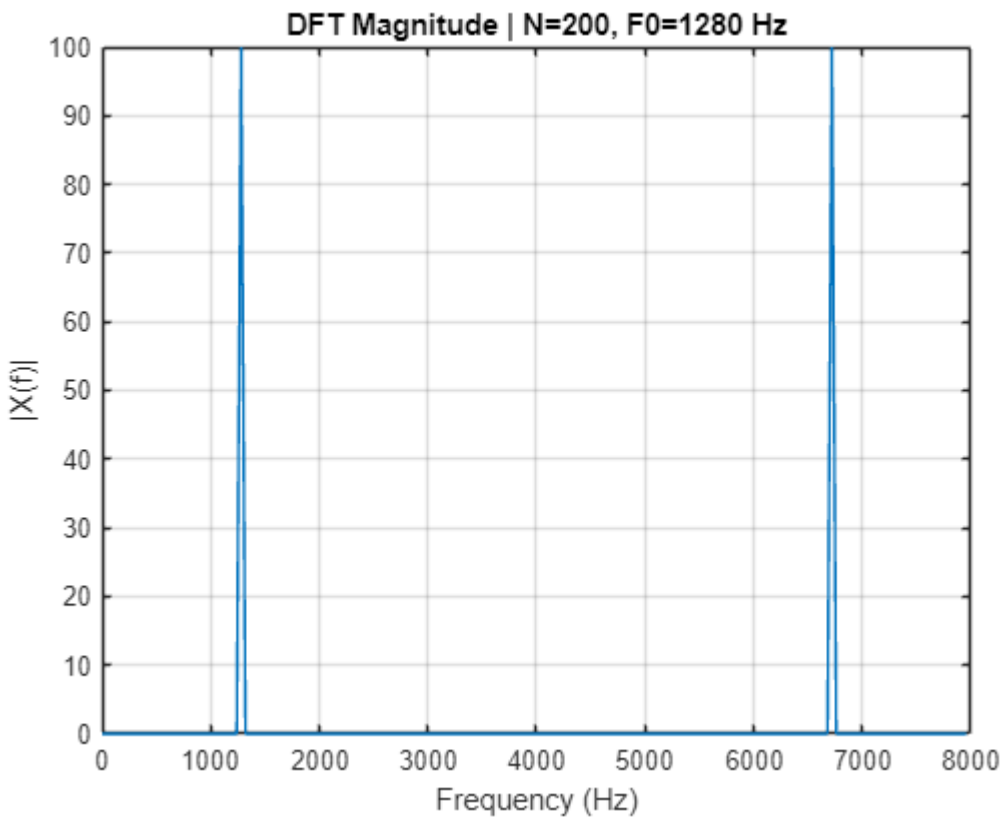
```
F0 = 1280;
N = 200;
dF = Fs/N
```

```
dF = 40
```

```
n = 0:N-1;
x = cos(2*pi*F0*n*T);
X = fft(x)
```

```
X = 1×200 complex
10^2 ×
  -0.0000 + 0.0000i  -0.0000 - 0.0000i   0.0000 - 0.0000i   0.0000 - 0.0000i ···
```

```
fv = (0:N-1)*dF;

figure;
plot(fv, abs(X))
title('DFT Magnitude | N=200, F0=1280 Hz')
xlabel('Frequency (Hz)')
ylabel('|X(f)|')
grid on
```



i. What are the frequency values of the two pulses? How is this related to F0?

F0 and Fs-F0

so 1280 and 6720 Hz

ii. What are the FFT indices of the two pulses? How is this related to F0 and Δ?

dF = Fs/N = 40

F0/dF = 32

Indices at 32 and N-32 = 168

iii. What is the amplitude of the two pulses? How is this related to N?

Amplitude = N/2 = 100

```
N = 100;
dF = Fs/N
```

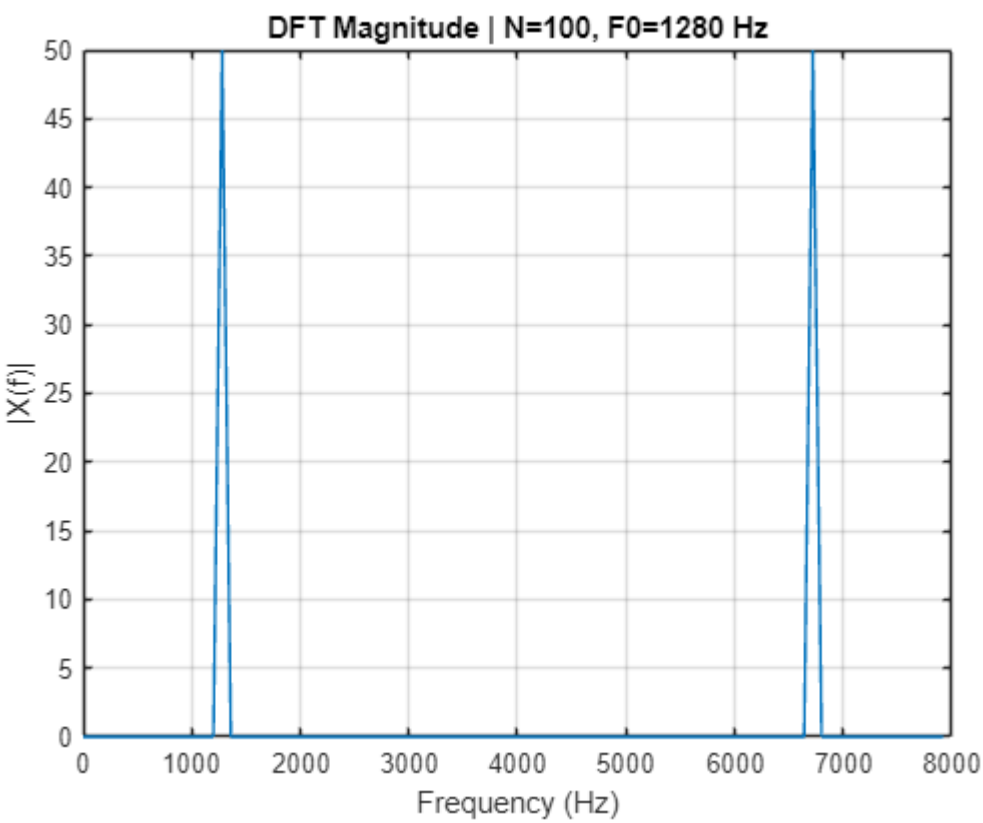
```
dF = 80
```

```
n = 0:N-1;
x = cos(2*pi*F0*n*T);
```

```
X = fft(x)
```

```
X = 1×100 complex
  -0.0000 + 0.0000i   0.0000 - 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i ···
```

```
fv = (0:N-1)*dF;

figure;
plot(fv, abs(X))
title('DFT Magnitude | N=100, F0=1280 Hz')
xlabel('Frequency (Hz)')
ylabel('|X(f)|')
grid on
```



i. What are the frequency values of the two pulses? How is this related to F0?

F0 and Fs-F0

so 1280 and 6720 Hz

ii. What are the FFT indices of the two pulses? How is this related to F0 and Δ?

dF = Fs/N = 80

F0/dF = 16

Indices at 16 and N-16 = 184

iii. What is the amplitude of the two pulses? How is this related to N?

Amplitude = N/2 = 50

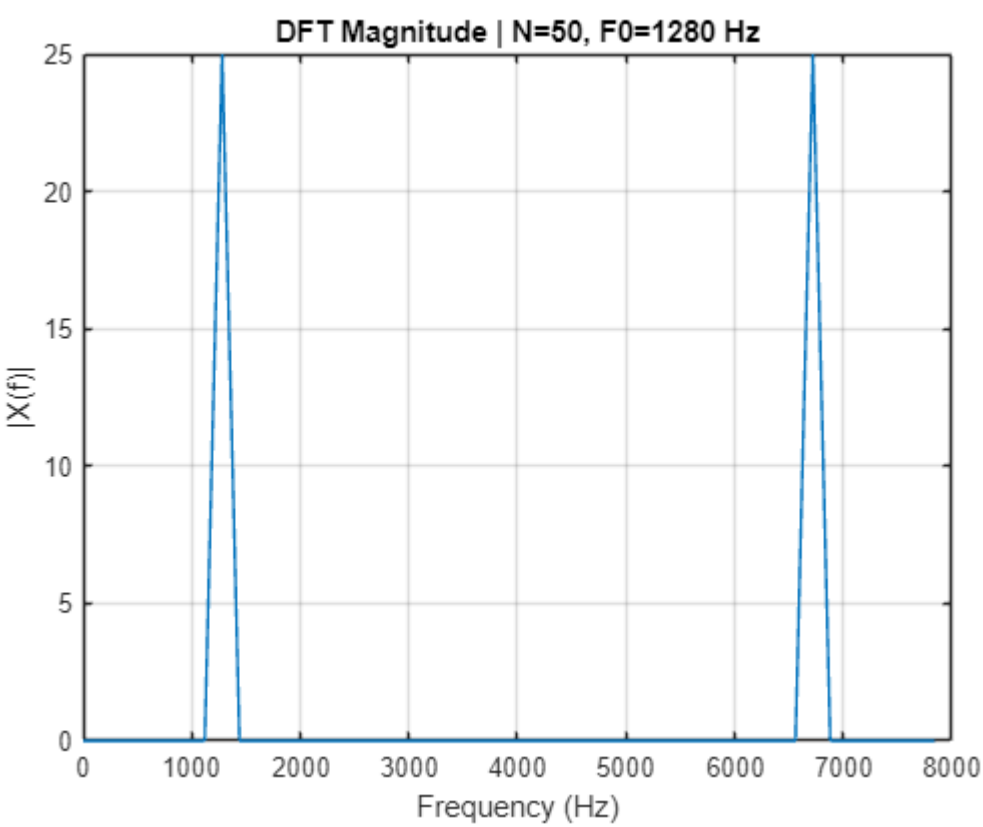
```
N = 50;
dF = Fs/N
```

```
dF = 160
```

```
n = 0:N-1;
x = cos(2*pi*F0*n*T);

X = fft(x)
```

```
X = 1×50 complex
  -0.0000 + 0.0000i  -0.0000 - 0.0000i  -0.0000 - 0.0000i  -0.0000 - 0.0000i ···
```

```
fv = (0:N-1)*dF;

figure;
plot(fv, abs(X))
title('DFT Magnitude | N=50, F0=1280 Hz')
xlabel('Frequency (Hz)')
ylabel('|X(f)|')
grid on
```



i. What are the frequency values of the two pulses? How is this related to F0?

F0 and Fs-F0

so 1280 and 6720 Hz

ii. What are the FFT indices of the two pulses? How is this related to F0 and Δ?

dF = Fs/N = 160

F0/dF = 8

Indices at 8 and N-32 = 192

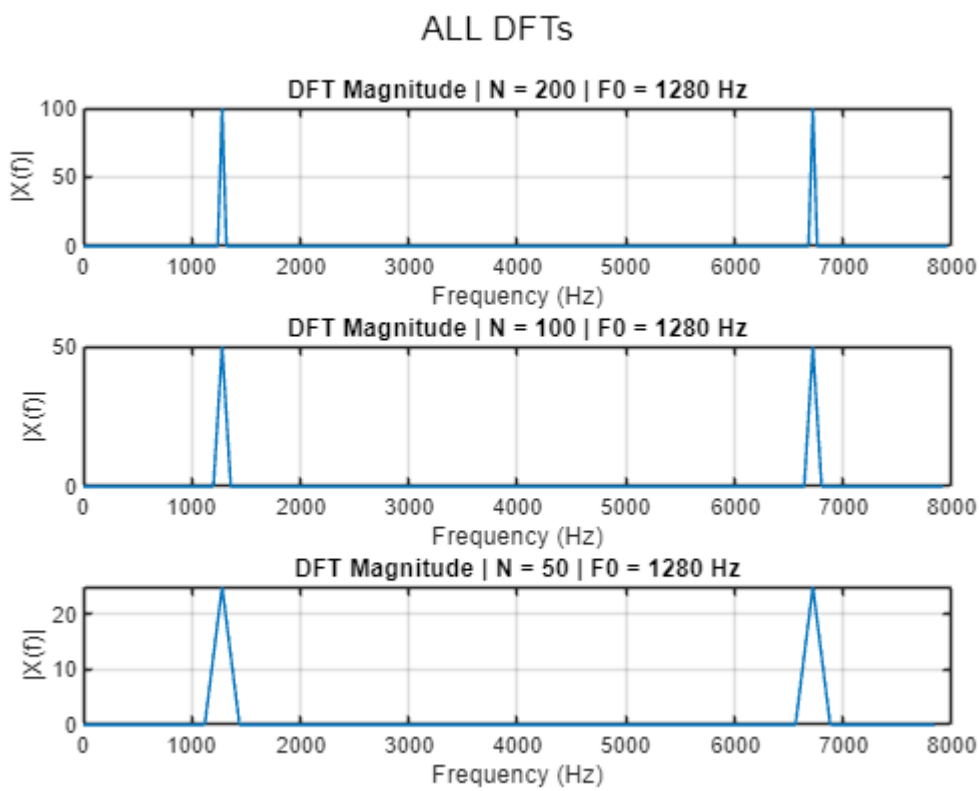
iii. What is the amplitude of the two pulses? How is this related to N?

Amplitude = N/2 = 25

```
F0 = 1280;

figure;
for idx = 1:3
    N_list = [200, 100, 50];
    N = N_list(idx);
    dF = Fs/N
    n = 0:N-1;
    x = cos(2*pi*F0*n*T);
    X = fft(x);
    fv = (0:N-1)*dF;

    subplot(3,1,idx)
    plot(fv, abs(X))
    title(['DFT Magnitude | N = ', num2str(N), ' | F0 = 1280 Hz'])
    xlabel('Frequency (Hz)')
    ylabel('|X(f)|')
    grid on
end
```

```
dF = 40
dF = 80
dF = 160
```

```
sgtitle('ALL DFTs')
```



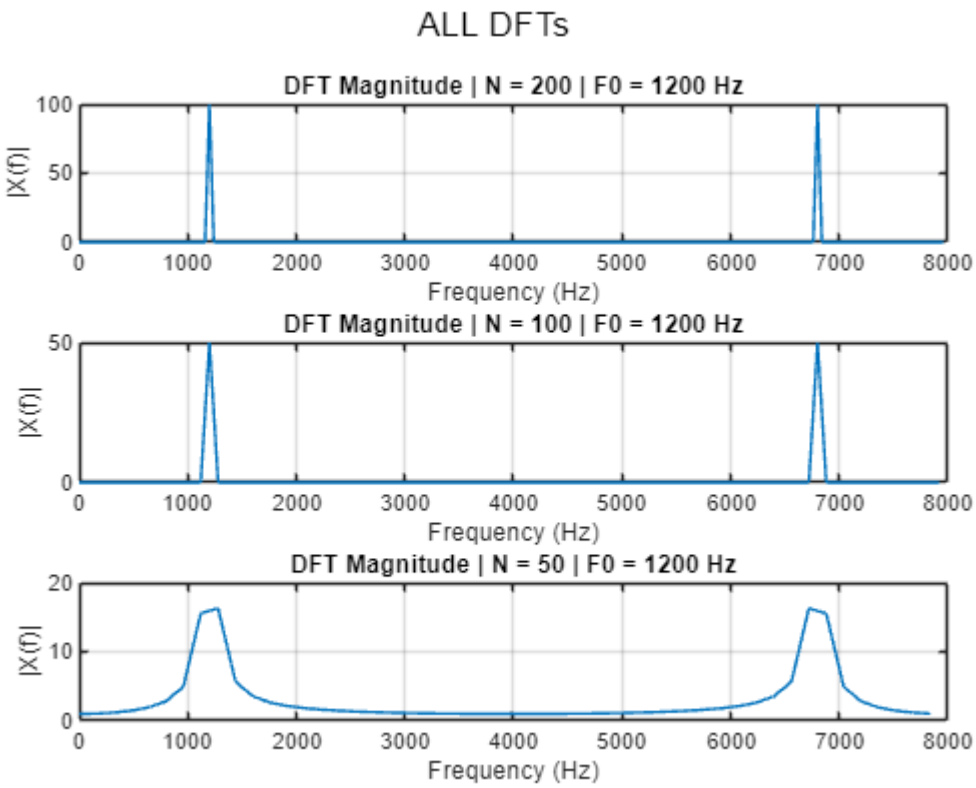
SECTION 3

```
F0 = 1200;

figure;
for idx = 1:3
    N_list = [200, 100, 50];
    N = N_list(idx);
    dF = Fs/N
    n = 0:N-1;
    x = cos(2*pi*F0*n*T);
    X = fft(x);
    fv = (0:N-1)*dF;

    subplot(3,1,idx)
    plot(fv, abs(X))
    title(['DFT Magnitude | N = ', num2str(N), ' | F0 = ', num2str(F0), ' Hz'])
    xlabel('Frequency (Hz)')
    ylabel('|X(f)|')
    grid on
end
```

```
dF = 40
dF = 80
```

```
dF = 160
```

```
sgtitle('ALL DFTs')
```



Do all three plots have only two pulses? Explain any difference in the plots of the DFTs compared to previous results.

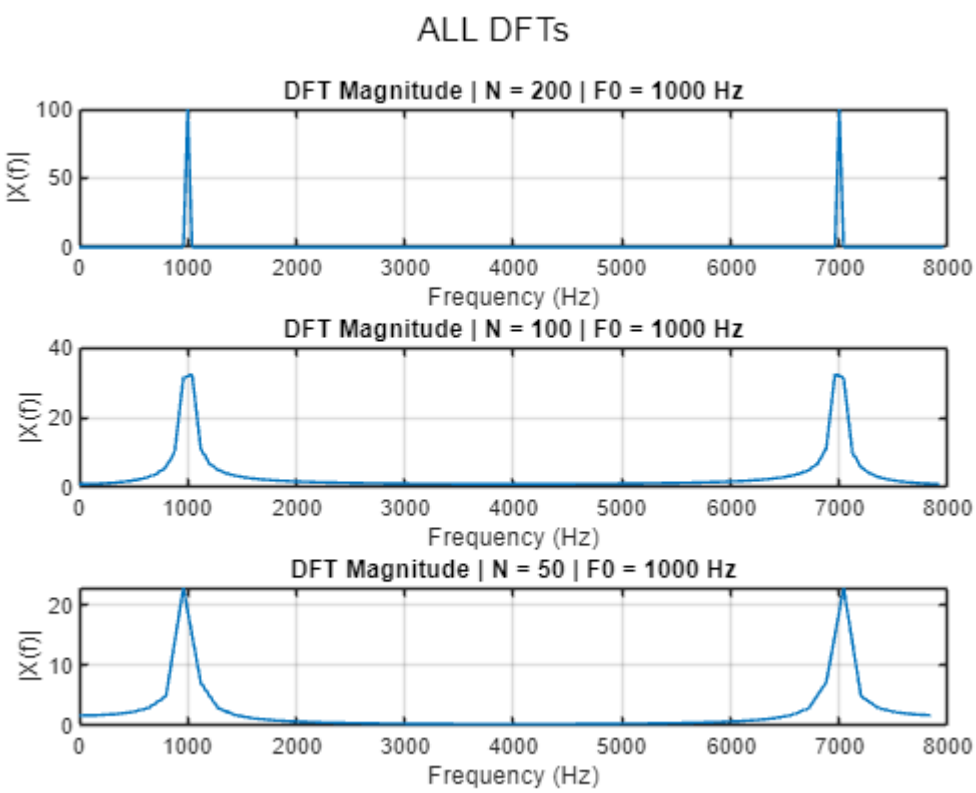
They do, but the N = 50 plot breaks and isn't spiky. This is likely because F0 isn't cleanly divisible by dF (160).

```
F0 = 1000;

figure;
for idx = 1:3
    N_list = [200, 100, 50];
    N = N_list(idx);
    dF = Fs/N
    n = 0:N-1;
    x = cos(2*pi*F0*n*T);
    X = fft(x);
    fv = (0:N-1)*dF;

    subplot(3,1,idx)
    plot(fv, abs(X))
```

```
    title(['DFT Magnitude | N = ', num2str(N), ' | F0 = ', num2str(F0), ' Hz'])
    xlabel('Frequency (Hz)')
    ylabel('|X(f)|')
    grid on
end
```

```
dF = 40
dF = 80
dF = 160
```

```
sgtitle('ALL DFTs')
```



Do all three plots have only two pulses? Explain any difference in the plots of the DFTs compared to previous results.

Now only N=200 has our expected behavior. Still has two pulses for all, but for N=100 and N=50, dF won't be a factor of 1000, thus our DFT breaks.

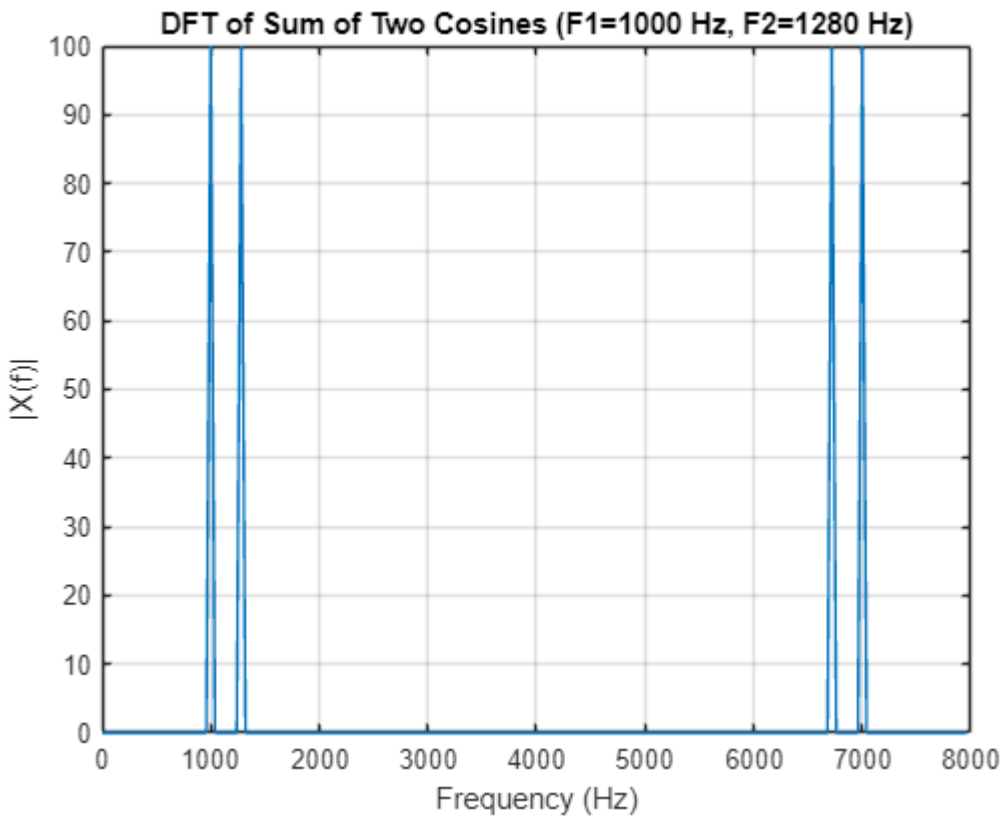
SECTION 5

```
N = 200;

F1 = 1000;
```

```
F2 = 1280;

dF=Fs/N
```

```
dF = 40
```

```
n = 0:N-1;
x = cos(2*pi*F1*n*T) + cos(2*pi*F2*n*T);

X = fft(x);
fv = (0:N-1)*dF;

figure;
plot(fv, abs(X))
title('DFT of Sum of Two Cosines (F1=1000 Hz, F2=1280 Hz)')
xlabel('Frequency (Hz)')
ylabel('|X(f)|')
grid on
```



Explain the DFT plot in terms of the previously created DFT plots. Are the two frequencies easily distinguished? Include this plot with your prelab.

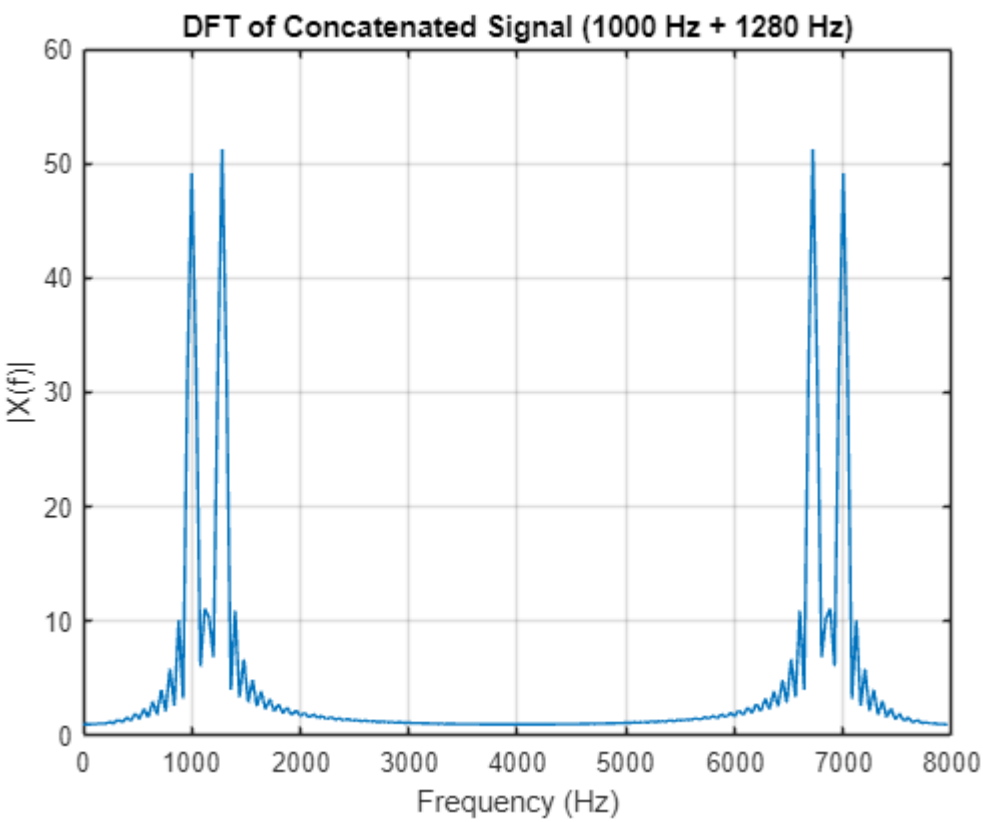
These frequencies are easily distinguishable and we can see that the DFT just added the two frequencies together. It looks like we just took the two plots and combined them.

SECTION 6

```
N = 100;

n1 = 0:N-1;
x1 = cos(2*pi*1000*n1*T);
x2 = cos(2*pi*1280*n1*T);
x = [x1, x2];

dF = Fs/(2*N)
```

```
dF = 40
```

```
X = fft(x);
fv = (0:2*N-1)*dF;

figure;
plot(fv, abs(X))
title('DFT of Concatenated Signal (1000 Hz + 1280 Hz)')
xlabel('Frequency (Hz)')
ylabel('|X(f)|')
grid on
```



Explain the DFT plot in terms of the previously created DFT plots for N = 100 and N = 200. Include this plot with your prelab.

Lots of leakage. Dividing the N/2 and then multiplying it by two after creating our signals and concatenating them didn't equate to adding them. We see a peak amplitude of ~100/2 and some ripples down low.